AdventureWorks Sales Analysis

Combining SQL and Python in an End-to-End Analytical Workflow

Extracting relational sales data with SQL and analyzing it with Python

---

By

Ahmed Nassr-Eldin El-Sayed

For

NTI Data Analytics Summer Training

Date

28/08/2026

GitHub Repository: https://github.com/ahmed-nassr/adventure-works-sales-analysis

SCOPE

Objective and Analytical Scope

Project goal: Demonstrate how SQL and Python complement each other in a structured, end-to-end data-analysis workflow applied to the AdventureWorks relational database.

Four Analytical Areas

01

Sales Performance Over Time

02

Product and Category Performance

03

Customer Contribution and Repeat Purchasing

04

Order Quantities and Their Relationship with Order Value

Principal Questions

- How does total order value change over time?
- Which products and categories contribute the most value?
- How concentrated is customer purchasing activity?
- How frequently do customers place multiple orders?
- How does order quantity relate to total order value?

METHODOLOGY

SQL × Python Workflow

End-to-End Pipeline

SQL Server / AdventureWorks

Source relational database containing transactional sales and product data

Table Selection and Joins

Cross-table joins connecting orders, details, products, subcategories, and categories

Python Connection via pyodbc

Query results loaded directly into pandas DataFrames for downstream processing

Analysis and Visualization

Validation, aggregation, derived metrics, and exploratory visual analysis

Principal Tables

Sales.SalesOrderHeader

Order-level records including dates, customers, and total amounts

Sales.SalesOrderDetail

Line-level detail including products, quantities, and line totals

Production.Product

Joined with subcategory and category tables for hierarchical classification

Division of Responsibilities

SQL

Connection, table selection, relational joins, data retrieval

Python

Validation, aggregation, derived metrics, EDA, visualization

PREPARATION

Data Validation and Analytical Preparation

Preparation Workflow

1. **Preview and Type Check**

   Extracted tables previewed and data types confirmed before aggregation

2. **Missing Values and Statistics**

   Reviewed null presence and descriptive statistics across all tables

3. **Order-Level Aggregation**

   Line-level quantities and values rolled up to order and product levels

4. **Category Linkage**

   Product performance connected with names, subcategories, and categories

Dataset Scale

30k+

Total Orders

504

Catalogue Products

52.8%

Products in Orders

266 of 504 appeared in recorded orders

19k+

Unique Customers

39%

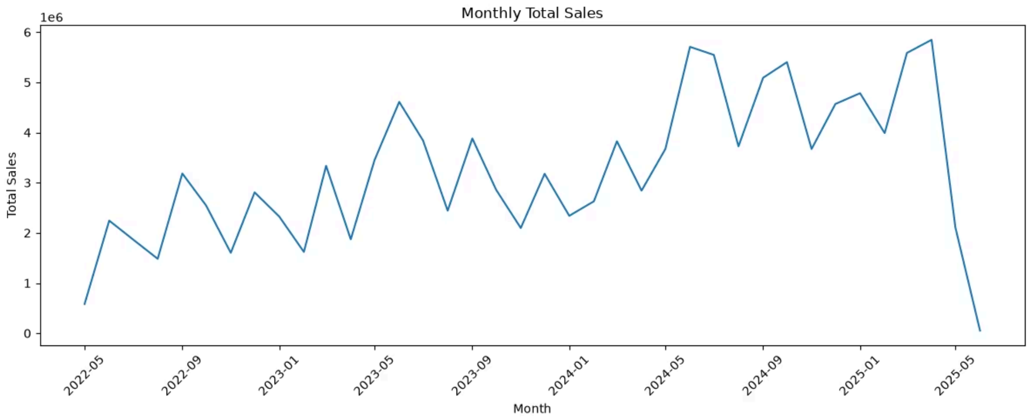
Repeat Purchasers

Customers with $\geqslant$2 recorded orders

ANALYSIS 1 OF 4

Total Order Value Over Time

Metric definition: TotalDue includes subtotal, tax, and freight — it is not a pure measure of product sales revenue.

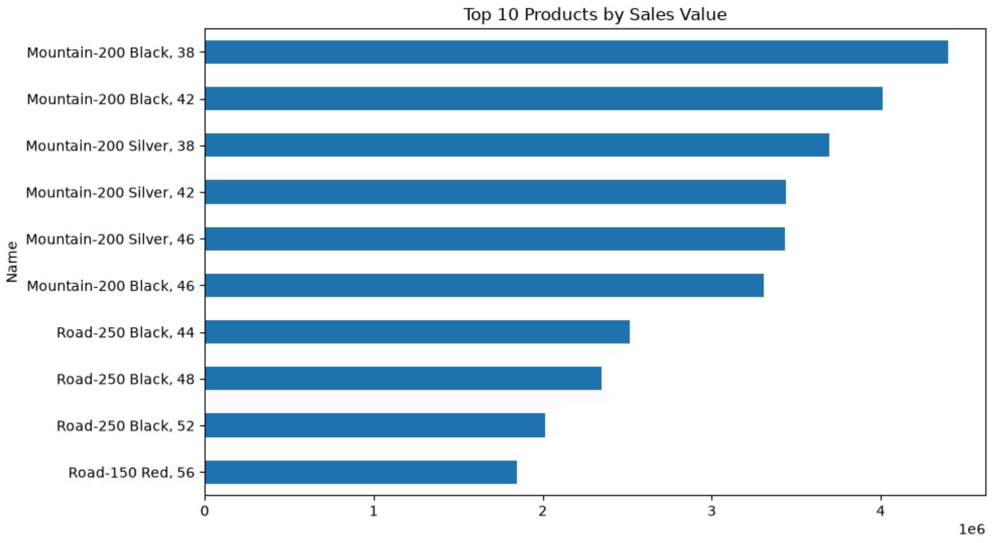


Observations

- Total order value exhibits **considerable month-to-month variation** throughout the observation period
- A **general upward trend** is visible in the middle period, with notable peaks around mid-2014
- A **sharp decline** is visible at the end of the observation period — cause is not determinable from this dataset alone

Product Performance

Product value is highly skewed. A logarithmic transformation was applied to reveal differences among lower- and middle-value products.



Key Insights

Leading Product

Mountain-200 Black (size 38) leads with the highest recorded product value in the dataset

Dominant Segment

The top 10 products are dominated by **Mountain and Road bike models**

Value Calculation

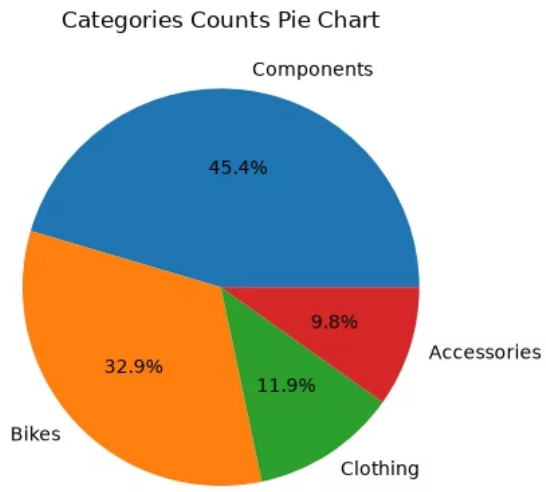
Product value = sum of LineTotal across all orders; distinct from unit count or order frequency

Price Effect

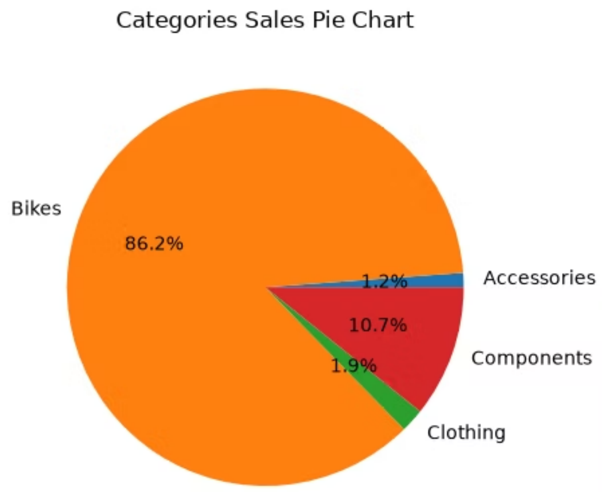
Value ranking differs from frequency ranking due to significant **variation in unit prices**

Category Composition vs. Value Contribution

Catalogue Distribution



Value Contribution



Key Mismatch:
Catalogue Share ≠ Value Share

Bikes

32.9% of catalogue products

→ **86.2%** of recorded value

Components

45.4% of catalogue products

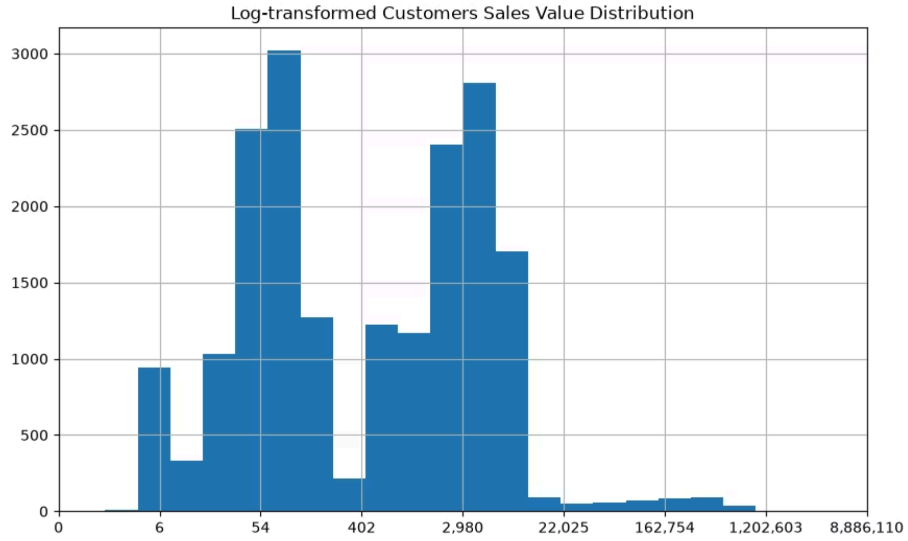
→ **10.7%** of recorded value

Takeaway

A category's share of the product catalogue does not determine its share of total order value

ANALYSIS 4 OF 4 — PART A

Customer Contribution and Repeat Purchasing



19,119

Unique Customers

Distribution Observations

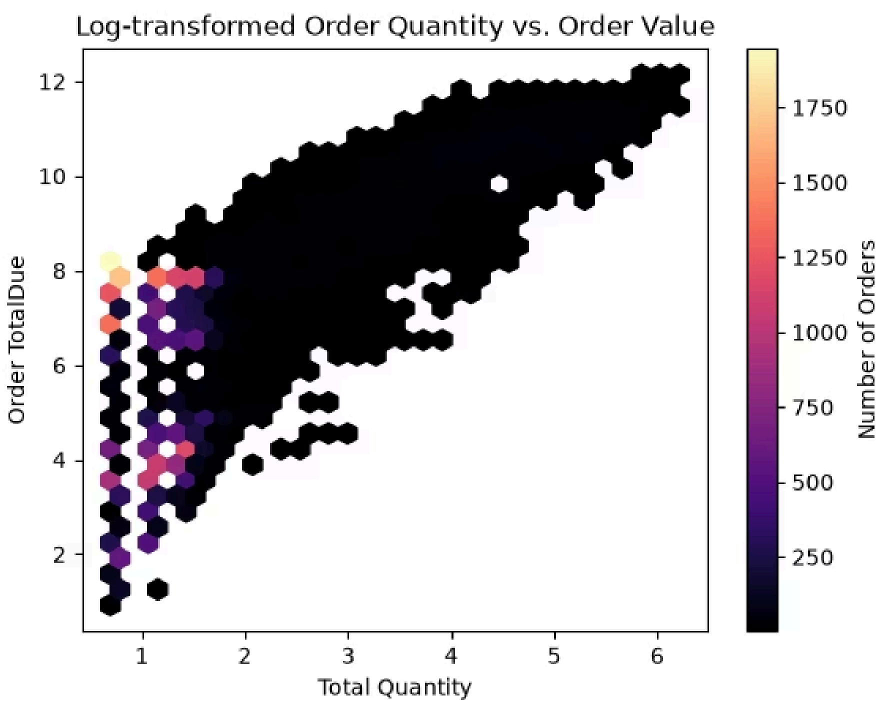
Customer total order value is **highly skewed** — most customers contribute relatively little in absolute terms

The log-transformed distribution shows **multiple concentrations**, but further segmentation would require additional customer attributes before drawing conclusions

The **39% repeat-purchase rate** is defined as customers with two or more recorded orders in the dataset; it does not imply loyalty behavior beyond what the data directly shows

ANALYSIS 4 OF 4 — PART B

Order Quantities and Order Value



Key Observations

Positive Relationship

Orders containing more units generally have greater total value — a positive association is visible in the hexbin density

Price Variation Effect

Quantity alone does not determine order value; significant variation in product prices moderates the relationship

Visualization Choices

Hexbin Density

Handles overlapping observations at scale; color intensity represents concentration of orders in each region

Dual Log Transformation

Both axes log-transformed to display the dense majority and extreme values inthe same view

SUMMARY

Conclusions and Learning Outcomes

Analytical Observations

Temporal Variation

Total order value changes considerably over time with visible trend and month-to-month variation

Partial Product Coverage

Only 266 of 504 catalogue products (52.8%) appeared in recorded orders

Bike Value Dominance

Bikes represent 32.9% of catalogue but account for **86.2%** of recorded order value

Skewed Customer Value

Customer purchasing value is strongly skewed; 39% of customers placed ⩾2 recorded orders

Quantity–Value Relationship

Positive association between order quantity and total value, moderated by product pricing

Technical Learning Outcomes

01

pyodbc Connection

Connecting Python to SQL Server and executing queries programmatically

02

SQL Joins

Retrieving and joining relational data across multiple tables

03

pandas Integration

Moving database results into DataFrames for flexible downstream analysis

04

Multi-Level Metrics

Validating tables and creating order-, product-, category-, and customer-level aggregates

05

Visualization Selection

Selecting appropriate charts for skewed distributions and overlapping high-density data

> **SQL provided structured access to relational business data, while Python provided the flexibility needed to validate, analyze, and visualize it.**

View the complete notebook and analysis on GitHub: https://github.com/ahmed-nassr/adventure-works-sales-analysis